Selecting a Robust LLM-Era Base Model for Recommendation Systems

Detailed explanation of 10 recent papers, our current project, and the model-to-paper mapping

Executive decision

We should retain SASRec as the conventional baseline and develop an LLM-enhanced dense sequential recommender as our main base model. We will use LLMs selectively for semantic item/user representations and reasoning distillation, while the final student remains a compact Transformer/SASRec-style encoder that ranks real products from the Amazon Beauty catalog. This gives us one shared backbone for centralized, federated, on-device, and differentially private experiments.

How this extends our current project

Our current project already has the right foundation: Amazon Beauty reviews and metadata, chronological next-product prediction, user-level client partitioning, and planned centralized, federated, compressed/on-device, and privacy variants. The research review changes the base-model choice, not the project goal. We keep SASRec as Model 0 and use the recent papers to design a modern semantic base and its practical extensions.

Current project versus updated design

Existing project	What we keep	What the research adds
Amazon Beauty ETL	Reviews, ratings, timestamps, product IDs, metadata, user partitions	Semantic item/user representations and cold-start tests
Next-item task	Chronological sequence construction and candidate ranking	Long-tail, sparse-user, diversity, and coverage analysis
SASRec baseline	ID-based sequential control	Fair reference for every LLM-era improvement
Four model variants	Centralized, federated, on-device, and private models	One shared dense student instead of unrelated architectures
Privacy objective	Federated learning plus differential privacy	Communication accounting and formal epsilon/delta reporting

Selection criteria

• Must model chronological behavior, not only text similarity.

• Must use semantic metadata without losing collaborative signals.

• Must be compatible with user-level federation and DP training.

• Must be compressible and measurable on resource-constrained devices.

• Must be reproducible on public Amazon-style data.

Evidence status

The papers below are treated as 2026 research because the current date is October 2026. ACL entries are peer-reviewed conference publications. LUMOS and GenRec are recent preprint/industry sources, and OD-LLM is retained with its WSDM 2026 paper/preprint status. Paper-reported numbers are not treated as our results until independently reproduced.

1. RecPO - What Makes LLMs Effective Sequential Recommenders? A Study on Preference Intensity and Temporal Context

Publication status: ACL 2026
What the paper contributes: The paper argues that ordinary preference optimization treats feedback as a simple binary choice, even though real users express different strengths of preference and their recent behavior may be more important than older behavior. RecPO combines explicit signals such as ratings with implicit interactions and builds adaptive reward margins using both preference intensity and recency. For our Amazon Beauty data, this is useful because ratings and timestamps are available. We can borrow its idea as an optional preference-weighting or time-decay module on top of the base sequence encoder. We should not use RecPO as the complete common backbone because its main contribution is preference alignment, not federated or on-device deployment.
Source: https://aclanthology.org/2026.acl-long.656/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

2. UCGRec - UCGRec: User-Centric Graph Learning for LLM-based Sequential Recommendation

Publication status: Findings of ACL 2026
What the paper contributes: UCGRec addresses the problem that a user may have several interests at the same time. Instead of passing isolated preference signals to an LLM, it organizes user-relevant information into a user-centric graph and injects graph knowledge into the LLM through end-to-end training. The paper is valuable for understanding how long-term interests, short-term intent, and collaborative relationships can be combined. For our team, it is an advanced optional extension rather than the first implementation. A graph may improve personalization, but constructing and protecting user graphs on federated clients would increase memory, communication, and privacy complexity.
Source: https://aclanthology.org/2026.findings-acl.175/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

What this pair contributes to our decision

These papers should be read together as design evidence, not as competing complete systems. We use their compatible ideas selectively and keep the evaluation task, data split, and catalog-ranking output consistent across our models.

3. HSUGA - HSUGA: LLM-Enhanced Recommendation with Hierarchical Semantic Understanding and Group-Aware Alignment

Publication status: Findings of ACL 2026
What the paper contributes: HSUGA focuses on two weaknesses of LLM-enhanced sequential recommendation. Long histories are difficult for an LLM to summarize reliably, and every user should not receive the same strength of semantic guidance. The paper hierarchically mines preferences from stages of the interaction history and changes the alignment strength according to user activity. This is relevant to Amazon Beauty because some users have many reviews while others have very short histories. We can borrow the idea for a sparse-user module: use stronger semantic guidance for users with limited interactions and rely more on collaborative sequence signals for active users.
Source: https://aclanthology.org/2026.findings-acl.726/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

4. GRAM / TIDs - Unleashing the Native Recommendation Potential: LLM-Based Generative Recommendation via Structured Term Identifiers

Publication status: Findings of ACL 2026
What the paper contributes: This paper tackles a central problem in generative recommendation: free-form text is difficult to map reliably to a real item, while numerical semantic IDs are not natural tokens for an LLM. The proposed Term IDs are structured sets of meaningful textual keywords associated with products. The framework jointly trains the model to identify these terms and recommend personalized items, while grounding the generated identifier back to the catalog. This is useful for understanding modern generative recommendation and item grounding. For our project, it should remain an optional research branch because full generative decoding is heavier than dense candidate ranking and is less convenient for federation and DP.
Source: https://aclanthology.org/2026.findings-acl.984/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

What this pair contributes to our decision

These papers should be read together as design evidence, not as competing complete systems. We use their compatible ideas selectively and keep the evaluation task, data split, and catalog-ranking output consistent across our models.

5. R2END - Distilling LLM Reasoning into Dense Encoders: Bridging the Accuracy-Efficiency Gap in Recommendation

Publication status: Findings of ACL 2026
What the paper contributes: R2END directly addresses the accuracy-efficiency problem in LLM recommenders. Rather than asking a student model to generate long reasoning text, it transfers the teacher LLM's useful reasoning into a dense vector space through semantic alignment. The final encoder can therefore preserve more semantic knowledge while avoiding autoregressive inference latency. This is the strongest base-model direction for our project. We can use a larger LLM offline to create profiles, preference signals, or reasoning targets, then train a compact SASRec/Transformer student. The student—not the large teacher—becomes the model used in centralized, federated, on-device, and DP experiments.
Source: https://aclanthology.org/2026.findings-acl.1130/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

6. BLaIR - Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders

Publication status: ACL 2026
What the paper contributes: BLaIR is especially important because it evaluates language models as semantic encoders for recommendation rather than assuming that general language or MTEB embedding scores are sufficient. It introduces a large Amazon Reviews 2023 resource and evaluates sequential recommendation, collaborative filtering, and product search. The practical lesson is that a text embedding that looks strong on a generic benchmark may not preserve the collaborative information needed for ranking products. We should therefore validate item embeddings on our recommendation task and compare ID-only, semantic-only, and fused representations. BLaIR guides our semantic input layer, not our entire sequential architecture.
Source: https://aclanthology.org/2026.acl-long.147/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

What this pair contributes to our decision

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7. Profile-then-Embedding - From ID to LLM: Rethinking Representation Learning for Recommendation

Publication status: ACL 2026
What the paper contributes: This paper challenges the assumption that ID representations and language-model representations must remain separate. Its Profile-then-Embedding framework first generates semantic profiles for users and items using bidirectional reasoning over interactions, and then encodes those profiles into task-aligned recommendation embeddings. The relevant idea for us is to separate expensive semantic reasoning from efficient recommendation inference. We can prepare item profiles offline from Amazon metadata and reviews, optionally create user profiles from local histories, and feed the resulting embeddings into the compact sequence encoder. This is a practical bridge between LLM knowledge and a deployable recommender.
Source: https://aclanthology.org/2026.acl-long.179/

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

8. LUMOS - Empowering Contrastive Federated Sequential Recommendation with LLMs

Publication status: ArXiv 2026 preprint
What the paper contributes: LUMOS is the closest paper to our federated model. It uses an LLM locally to generate three views of a user sequence: future-oriented trajectories, semantically equivalent paraphrases, and preference-inconsistent counterfactuals. These views are used in a tri-view contrastive objective so that each client can learn richer representations without sharing raw histories or auxiliary LLM parameters. We can borrow this idea after implementing ordinary FedAvg or FedProx. Each user remains a client, the compact student is aggregated, and local semantic views are used only to strengthen representation learning. We must measure the extra local computation and clearly label LUMOS as a preprint.
Source: https://arxiv.org/abs/2602.09306

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

What this pair contributes to our decision

These papers should be read together as design evidence, not as competing complete systems. We use their compatible ideas selectively and keep the evaluation task, data split, and catalog-ranking output consistent across our models.

9. OD-LLM - On-Device Large Language Models for Sequential Recommendation

Publication status: WSDM 2026 paper/preprint
What the paper contributes: OD-LLM focuses on the deployment problem created by large model size and computation. It combines low-rank structural compression using SVD, tokenization normalization, and progressive alignment to reduce model size while protecting recommendation quality. The reported direction is useful for our on-device model because it gives us concrete mechanisms to test rather than only saying that the model is “lightweight.” We can apply low-rank factorization, quantization, or student distillation to the selected semantic student and report memory, latency, throughput, energy, and accuracy loss. We should reproduce measurements on our actual target hardware instead of assuming paper-level efficiency.
Source: https://arxiv.org/abs/2601.09306

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

10. GenRec - GenRec: An LLM-Backed Recommendation Ranker at Netflix

Publication status: ArXiv 2026 industry preprint
What the paper contributes: GenRec provides industrial motivation rather than a directly reproducible open model. Netflix describes a two-phase process: first adapt an open-source or internal LLM to the catalog and member behavior, then post-train it with recommendation labels, reward signals, and business objectives. The system uses verbalized user histories and context and discusses cost-constrained serving. This supports our argument that product companies are exploring shared LLM backbones and context engineering. However, Netflix-scale data, internal models, and online A/B infrastructure are unavailable to us. We should cite GenRec as industry evidence, not claim that we reproduce Netflix’s ranker.
Source: https://arxiv.org/abs/2608.10257

Team takeaway: Treat this paper as either the source of a specific module or as evidence for a design decision; do not present its complete system as if it were directly reproduced unless the code, data, and training setup are independently matched.

Cross-paper conclusion

The papers collectively show a clear evolution from Transformer-based ID sequence modeling toward hybrid LLM recommenders. The LLM is most useful for semantic understanding, preference extraction, data augmentation, or reasoning distillation. The practical recommender should still use a grounded dense encoder because it is faster, easier to evaluate, and easier to adapt to federated learning, edge deployment, and formal privacy.

What no single paper solves

• No selected paper simultaneously provides semantic recommendation, federated learning, on-device serving, and formal differential privacy.

• LLM-generated profiles and synthetic sequences can introduce hallucination, popularity bias, or privacy leakage.

• Offline next-item metrics do not measure long-term satisfaction or online business impact.

• Preprint and industry claims must be labeled separately from peer-reviewed evidence.

Model-to-paper mapping for our team

Our model	Borrowed paper idea	What we will implement	Why it fits
Model 0: Centralized SASRec	SASRec/Transformer sequential recommendation	ID embeddings, chronological history, next-item ranking on Amazon Beauty.	Unchanged control baseline.
Model 1: LLM-enhanced base	R2END + BLaIR + Profile-then-Embedding	Offline semantic item/user representations, LLM-to-dense distillation, and fusion with compact SASRec/Transformer.	Best balance of novelty, accuracy, grounding, and efficiency.
Model 2: Federated	LUMOS	User-level clients, FedAvg/FedProx, optional future/paraphrase/counterfactual views generated locally.	Tests decentralized personalization and robustness.
Model 3: On-device	OD-LLM	Low-rank compression, quantization, or distillation applied to Model 1; measure actual device cost.	Tests practical edge inference.
Model 4: Differential privacy	DP-SGD / DP-FedAvg applied to Model 1 or Model 2	Gradient clipping plus calibrated Gaussian noise with epsilon/delta accounting.	Tests formal privacy rather than only local storage.

How to explain the borrowing to teammates

We are not copying five complete systems. We borrow one validated idea from each source and apply it to the same Amazon Beauty backbone: R2END supplies semantic distillation, BLaIR and Profile-then-Embedding guide the representation layer, LUMOS supplies federated sequence augmentation, OD-LLM supplies compression, and DP-SGD/DP-FedAvg supplies formal privacy accounting. This makes the comparison controlled and gives every model a clear research origin.

Implementation flow

Stage	Team action	Output
1. Baseline	Freeze ETL, chronological split, client partitions, and SASRec training.	Reproducible Model 0 and data statistics.
2. Semantic base	Compare ID-only, semantic-only, and fused embeddings; then distill teacher knowledge into the student.	Model 1 checkpoint and ablation table.
3. Federation	Run plain FedAvg/FedProx first; add LUMOS-inspired local views only after convergence is stable.	Model 2 plus rounds, bytes, drift, and quality.
4. Edge	Compress the Model 1 student and measure memory, latency, throughput, and energy.	Model 3 deployment profile.
5. Privacy	Add DP-SGD or DP-FedAvg after the non-private model is stable.	Model 4 privacy-utility curve.
6. Analysis	Test sparse users, long-tail items, cold-start products, noisy histories, and client heterogeneity.	Robustness and failure-case analysis.

Evaluation requirements

Category	Metrics to report	Interpretation
Accuracy	Recall@K, NDCG@K, HitRate@K, MRR	Does the model rank the next product correctly?
Personalization	Long-tail hit rate, coverage, diversity, cold-start performance	Does it help beyond popular products?
Federated	Rounds, bytes, client drift, convergence, client fairness	What is the decentralized cost and stability?
Efficiency	Parameters, peak memory, latency, throughput, energy	Can it run in a real-time or edge setting?
Privacy	Epsilon, delta, clipping norm, noise multiplier, attack risk	What formal protection is obtained and at what utility loss?

Final recommendation to present to the professor

We will retain SASRec as our conventional baseline and develop an LLM-enhanced dense sequential recommender as the main base model. We use LLMs for semantic representations and knowledge distillation rather than expensive free-form generation. The same student backbone will then be extended to federated, on-device, and differentially private settings. Our contribution is a controlled study of how modern LLM capabilities improve recommendation while preserving catalog grounding, deployment efficiency, decentralized training, and measurable privacy.

Implementation safeguards and claims

Area	What we must do	What we can claim
Data leakage	Use chronological splits and fit semantic preprocessing only on training information where user behavior is involved.	The comparison is temporally valid.
Semantic quality	Compare ID-only, semantic-only, and fused inputs; inspect incorrect or hallucinated profiles.	LLM semantics help only when they improve recommendation metrics.
Federation	Report client sizes, local epochs, rounds, aggregation, and non-IID behavior.	Federated training preserves raw-data locality under the chosen protocol.
Compression	Measure the same model before and after compression on the same hardware.	The compressed model has a measured efficiency/quality trade-off.
Privacy	Report clipping, noise, sampling, steps, epsilon, delta, and utility loss.	The DP model has a quantified formal privacy guarantee.
Industry relevance	Use GenRec only as industrial motivation, not as a reproducible baseline.	Our public-data experiment is inspired by product needs, not a Netflix reproduction.

Team discussion checklist

- Can every teammate explain what is borrowed from R2END, BLaIR, Profile-then-Embedding, LUMOS, and OD-LLM?
- Are all variants using the same task, split, candidate protocol, and core student architecture?
- Do we separate semantic improvement from federation, compression, and privacy effects?
- Do our tables report both accuracy and system cost?
- Are preprint findings clearly labeled and not presented as completed project results?

Expected limitations and mitigation

Limitation	How we handle it
Amazon Beauty is smaller than an industrial catalog.	Avoid direct claims about Netflix-scale performance; report the experiment as a public-data validation.
LLM semantic profiles may be noisy or hallucinated.	Use validation rules, ablations, item-grounding checks, and ID-only comparisons.
Federated clients are non-IID and may be sparse.	Report client-size distributions, convergence, drift, and performance for sparse users.
Privacy noise can reduce accuracy.	Show the full privacy-utility curve instead of reporting one selected epsilon only.
Offline metrics do not equal long-term satisfaction.	Include diversity and coverage, and state that online A/B testing is future work.

The final team message

Our project is a controlled study of how LLM capabilities can improve a practical sequential recommender. We are not claiming that a general-purpose LLM alone replaces a recommender system. We are testing where LLM knowledge helps, how much of it can be distilled into a compact model, and what accuracy, privacy, communication, and deployment costs remain when the same model is adapted to realistic product constraints.

What each model actually trains

Model	Training signal	Final output
SASRec baseline	Next-item cross-entropy or sampled-softmax objective over item IDs.	Catalog item ranking.
LLM-enhanced base	Same next-item objective plus semantic alignment/distillation loss.	Catalog ranking with semantic personalization.
Federated model	Local next-item and contrastive losses, followed by client update aggregation.	Global model without sharing raw histories.
On-device model	Task loss plus compression/distillation alignment during or after training.	Small model with measured latency and memory.
DP model	Clipped gradients plus Gaussian noise under DP-SGD or DP-FedAvg.	Private model with epsilon/delta accounting.

The recommendation target stays constant; semantic, decentralized, efficiency, and privacy mechanisms are changed one at a time. Each paper source is listed directly below its explanation, so a separate reference page is unnecessary.